

Powder Physics

A real-time 2D powder physics simulation focused on granular materials, cellular interactions, and emergent behavior.

How to Use

1. **Download** the [.jar release](#)
 2. **Extract** the downloaded folder
 3. **Run** the executable
 - If you encounter a "Permission Denied" error, run the following command to set the correct permissions:

```
bash chmod 700 <path to file>
```
-

Features

Grid-Based World

- The simulation runs on a 2D grid
- Each cell in the grid represents a single entity
- Designed for deterministic and scalable updates

Material Categories

Supports multiple material categories, each with unique behavior:

Solids

- Sand
- Stone
- Dirt

Liquids

- Water
- Oil
- Honey
- Lava
- Acid

Gases

- Smoke

Entity Composition System

- Each grid entity is built using composition
- Makes the system modular, scalable, and easy to extend
- New material types can be added without modifying existing logic

Physics & Movement

- Velocity-based movement for all entities
- Per-type collision checks and interaction rules
- Realistic material behavior determined on a type-by-type basis

User Controls

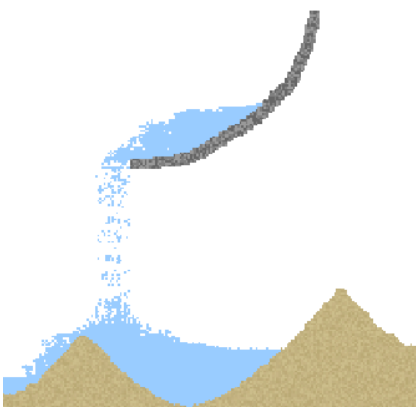
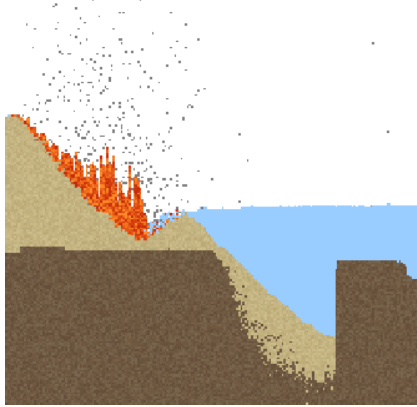
- Simple and intuitive menu for selecting material types
- Adjustable brush size for precise or large-scale placement
- Real-time interaction with the simulation

Multithreaded Simulation

- The grid is split into column-based chunks
- Multiple columns are processed simultaneously
- Improves performance and scalability on larger worlds

Pictures

Explore the simulation with realistic physics interactions.

Example 1	Example 2	Example 3																								
Brush Size ->  Erase <table><tr><td>Sand</td><td>Dirt</td><td>Stone</td><td>Water</td></tr><tr><td>Oil</td><td>Honey</td><td>Acid</td><td>Lava</td></tr></table> 	Sand	Dirt	Stone	Water	Oil	Honey	Acid	Lava	Brush Size ->  Erase <table><tr><td>Sand</td><td>Dirt</td><td>Stone</td><td>Water</td></tr><tr><td>Oil</td><td>Honey</td><td>Acid</td><td>Lava</td></tr></table> 	Sand	Dirt	Stone	Water	Oil	Honey	Acid	Lava	Brush Size ->  Erase <table><tr><td>Sand</td><td>Dirt</td><td>Stone</td><td>Water</td></tr><tr><td>Oil</td><td>Honey</td><td>Acid</td><td>Lava</td></tr></table> 	Sand	Dirt	Stone	Water	Oil	Honey	Acid	Lava
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